INCH-POUND

MIL-P-63140A (AR)
AMENDMENT 4
9 May 1990
SUPERSEDING
AMENDMENT 3
5 January 1989

MILITARY SPECIFICATION

PROJECTILE, 8-INCH, HE, RA, M650
DELAY ASSEMBLY FOR

This amendment forms a part of Military Specification MIL-P-63140A (AR), dated 24 March 1980 and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1 Under Military Specifications add the following:

"MIL-M-382C - Magnesium Powder (for use in ammunition)

MIL-P-217 - Potassium Perchlorate

MIL-T-48140 - Tungsten Powders (for pyrotechnics)"

2.1 Under Military Standards, change:

"MIL-STD-453 to MIL-STD-00453"

Add the following Military Standard:

"MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification

MIL-STD-414 - Sampling Prodcedures and Tables for Inspection by Variables for Percent Defective"

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2.1 Inspection Equipment Drawings:

Delete: "9325984 - Envelope Drawing (Porosity Test)" and substitute: "9246420 - Envelope Drawing (Porosity Test)"

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Delete: "9325985 - Nitrogen Test Assembly" and
substitute: "9326889 - Nitrogen Test Assembly"

* 2.2 Add the following.

"2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 430 - Particle Size Distribution of Refractory Metal-Type Powders by Turbidimetry

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)"

* 2.3 Add the following:

"2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

3.3.1 Delete in its entirety.

3.4 Delete in its entirety and substitute the following:

"3.4 Crimp strength. The washer crimp shall be capable of withstanding a 225 pound minimum load when tested as specified in 4.4.3.6."

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3.7 Flight test. Delete in its entirety and substitute the following:

"3.7 Flight test.

3.7.1 Delay chemicals acceptance.

3.7.1.1 Pilot mixtures. Small batches from each new lot of chemicals shall be prepared as shown in Table I.

TABLE I. Pilot mixtures.

Batch #	%W	%$BaCrO_4$	%$KClO_4$
1	41.00 ± .10	44.50 ± .10	14.50 ± .10
2	43.00 ± .10	42.50 ± .10	14.50 ± .10
3	45.00 ± .10	40.50 ± .10	14.50 ± .10
4	47.00 ± .10	38.50 ± .10	14.50 ± .10
5	49.00 ± .10	36.50 ± .10	14.50 ± .10

% VAAR shall be kept constant at approx. 1%.

3.7.1.2 Testing. Ten delay assemblies from each batch shall be submitted for ballistic firing. After conditioning at 70°F for a period of 24 hours minimum, the delays shall be flight tested in accordance with 4.5.3.1.1.

3.7.1.3 Tungsten curve. The ballistic results shall be plotted against the percent tungsten. From this curve the optimum tungsten percent will be determined. The chemical composition is optimized at a delay time of 7.8 seconds.

3.7.1.4 Usage. The results from paragraphs 3.7.1.1-3.7.1.3 shall be used in subsequent delay composition preparation.

3.7.2 Delay composition acceptance.

3.7.2.1 Plus 70°F. After conditioning at +70°F for a period of 24 hours min., the delay shall be flight tested at a proving ground as specified in 4.4.3.7.1.1. The average delay time shall be 7.8 plus or minus .75 sec., and the standard deviation of the delay times shall not exceed 0.5 sec.

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3.7.2.2 Plus 145°F. After conditioning at +145°F for a period of 24 hours min., the delay shall be flight tested at a proving ground as specified in 4.4.3.7.1.2. The average delay time shall be 7.05 plus or minus .75 sec. and the standard deviation of the delay times shall not exceed 0.5 sec.

3.7.3 Delay assembly acceptance.

3.7.3.1 Plus 145°F. After conditioning at +145°F for a period of 24 hours min., the delay assembly shall be flight tested at a proving ground (see 4.4.3.7.2.1). The average delay time shall be 7.05 seconds plus or minus .75 seconds, and the standard deviation shall be no greater than .5 sec.

3.7.3.2 Minus 50°F. After conditioning at minus 50°F for a period of 24 hours min., the delay assembly shall be flight tested at a proving ground (see 4.4.3.7.2.2). The average delay time shall be 9.30 seconds plus or minus 1.50 seconds, and the standard deviation shall be no greater than .8 sec.

3.7.3.3 Plus 70°F. After conditioning at +70°F for a period of 24 hours min., the delay assembly shall be flight tested at a proving ground (see 4.4.3.7.2.3). The average delay time shall be 7.8 seconds plus or minus 0.90 seconds and the standard deviation of the delay times shall be no greater than 0.5 seconds."

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Add the following new subparagraph:

"3.10 Process control for delay increment consolidation. A process control description shall be developed for delay increment consolidation, and submitted for government approval. The description shall include all elements of press setup, operation, calibration, maintenance, and operator instruction required to assure that the required number of delay increments are individually consolidated at the required pressure and for the required minimum consolidation time. The process controls shall include verification that the required pressure and dwell are met at the extremes of the press consolidation range (Section 4.4.3.1 and 6.12)."

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* 3.11 Add 3.11 as follows:

"3.11 Particle size.

3.11.1 Tungsten particle size. The tungsten shall be supplied de-agglomerated and shall meet all requirements of MIL-T-48140 except that particle size distribution shall be in accordance with Table II.

TABLE II. Particle ranges.

Particle Ranges		Microns		Percent	
Greater Than or Equal to		But Less Than		Min	Max
0		1		0	5
1		2		1	11
2		3		10	20
3		4		15	27
4		5		15	27
5		6		10	20
6		7		4	12
7		25		Balance	

In addition, the tungsten shall have a median particle size (50% point) of 5.0 to 5.8 microns. Ninety percent of the particles shall be 10 microns or smaller. Sampling shall be as specified in 4.4.3.11 and testing shall be as specified in 4.5.6.

3.11.2 Potassium perchlorate particle size. In addition to the requirements of MIL-P-217, the particle size at the 50% level shall be 25 $\pm$ 3 microns. Sampling shall be as specified in 4.4.3.11 and testing shall be specified in 4.5.6."

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4.4.2.4 Delete Minor Defect 201.

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4.4.2.5 Add the following Minor Defect "203 Distance tube end surface to inside bottom of igniter plug, min. -, 0.65%, 3.2, Gage".

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* 4.4.3.1 Delete in its entirety and substitute the following:

"4.4.3.1 Consolidation load. Major Defect. The contractor shall provide adequate controls to assure the compliance of each lot with the specified load requirement and dwell time. Proper calibration of each consolidation machine shall be verified at the beginning, middle and end of each production shift. Dwell time and consolidation load shall be measured with a stopwatch and a load cell. If any consolidation machine fails verification for dwell time, consolidation load or for not maintaining proper load through the dwell time, all the delays manufactured on that machine since the last acceptable verification shall be rejected. In addition, a warning device indicating failure of dwell time and failure of consolidation load shall be assembled to each consolidation machine. The warning device or mechanism shall provide an easily detected signal to the machine operator. If during loading of the delay the consolidation load or dwell time requirements are not met, that delay shall be rejected."

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Add the following new paragraph:

4.4.3.7.1.1 Delete in its entirety and substitute the following:

"4.4.3.7.1.1 Plus 70°F Major Defect. The acceptance criteria shall be in accordance with MIL-STD-414, Section B, Table B3, code letter F (10 samples), and an AQL of 6.5%. Each lot of delay composition shall be equally divided into ten sub-lots. The ten sample delay assemblies shall be submitted to the government for flight testing. If the requirement of paragraph 3.7.2.1 is not met--Major Defect, or if the delay assembly fails to function-major defect, or functions in bore-critical defect, the lot shall be rejected. The test shall be conducted in accordance with paragraph 4.5.3.1. A minimum of 8 radar delay times shall be required to determine delay time acceptance."

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Add the following new paragraph:

"4.4.3.7.1.2 Plus 145°F - Major defect. Each lot of delay composition shall be equally divided into ten sub-lots. One delay assembly shall be loaded from each sub-lot. The ten sample delay assemblies shall be submitted to the Government for flight testing. If the requirement of Paragraph 3.7.2.2 is not met - Major defect, or if the delay assembly fails to ignite - Major defect, or ignites in the tube - Critical defect, the delay assembly lot shall be rejected. The test shall be conducted in accordance with Paragraph 4.5.3.1.2. A minimum of 8 radar delay times shall be required to determine delay time acceptance."

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4.4.3.7.2.1 and 4.4.3.7.2.2 Delete in their entirety and substitute the following paragraphs:

"4.4.3.7.2.1 Plus 145°F. A sample of ten (10) loaded delay assemblies shall be selected at random and forwarded to a Government proving ground for firing. If the sample fails to meet the requirements of 3.7.3.1 - Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Major defect, the lot shall be rejected. A minimum of 8 delay times shall be required to determine delay time acceptance. The test shall be conducted in accordance with 4.5.3.2.1.

4.4.3.7.2.2 Minus 50°F. A sample of ten (10) loaded delay assemblies shall be selected at random and forwarded to a Government proving ground for firing. If the sample fails to meet the requirements of 3.7.3.2 - Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Major defect, the lot shall be rejected. A minimum of 8 delay times shall be required to determine delay time acceptance. The test shall be conducted in accordance with 4.5.3.2.2".

4.4.3.7.2.3 Delete in its entirety and substitute the following:

"4.4.3.7.2.3 Plus 70°F. The acceptance criteria shall be in accordance with MIL-STD-414, Section B, Table B3, code letter F (10 samples) and an AQL of 10%. A sample of ten (10) delay assemblies shall be selected at random and forwarded to a government proving ground. If the sample

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fails to meet the requirements of 3.7.3.3--Major defect, or if any delay fails to function, Major defect, or if any delay functions in bore--Critical defect, the lot shall be rejected. A minimum of 8 radar delay times shall be required to determine delay time acceptance."

4.4.3.8 Add the following at the end of the paragraph:

"Should the solids content be too high, additional solvent may be added, with agitation, until the solids content is acceptable. The VAAR solution will then be retested to assure compliance."

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* 4.4.3.11 Add as follows:

"4.4.3.11 Particle size. Prior to mixing of each composition lot, or pilot mixture, a sample of tungsten and of potassium perchlorate shall be tested for particle size from each shipping or storage container from which chemicals will be used in the mix. If the particle size of any sample tested fails to comply with the requirements of 3.11, chemical from that container shall not be used in the mix. Chemicals may be retested for use after application of suitable rework. A sample of 15 g minimum shall be taken from each individual container in such a manner that the sample is representative of the entire contents of the container. No lab milling, mulling, grinding or similar operation is permitted during sampling or preparation thereof."

4.5.1 Delete the first sentence and substitute the following:

"Each delay assembly shall be conditioned at plus 145°F and assembled to the porosity test equipment."

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4.5.2 Delete the last sentence and add the following at the end of the paragraph:

"In lieu of a penitrometer, a radiographic standard shall be manufactured to assure adequate sensitivity of the radiographic technique. A delay will be manufactured with a low density pocket in each of the two delay columns. Normal production materials shall be used to create the standard delay. In both columns the flash and first delay increment shall be pressed in correctly. In one column the second and third delay increments shall be placed in the column but not pressed, the fourth and fifth increments shall be pressed.

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In the other column the second increment will be correctly pressed, but the third and fourth increments will be loose loaded, the fifth increment will be pressed. Both columns will be back drilled, have igniter pressed in and have washers assembled. The delay manufactured as above will have a void between the first and second delay increments in one column and a void between the second and third increments in the other column. These voids must be clearly visible when the standard delay is radiographed. Using the standard delay, film sensitivity shall be checked on each piece of film used for delay acceptance. Photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.5.2.1."

Add the following new subparagraph:

"4.5.2.2 Operator qualification. For operation of radiograph equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II."

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* 4.5.6 Add the following new paragraphs:

"4.5.6 Particle size.

4.5.6.1 Particle size analyzer. Sedigraph particle analyzer 5000D or an approved equivalent instrument that uses a finely collimated beam of x-rays to measure particle concentration as a function of time is required. The model 5000D can be secured from Micromeritics Inst. Corp.

4.5.6.1.1 Dispersing fluid. The sedimentation fluid shall be Sedisperse A12, a product of Micromeritics, or an approved equivalent.

4.5.6.2 Test sample. Fill a 25 ml beaker with dispersing fluid and adequate sample powder for 100% recorder pen deflection when the beaker is placed in test position and the instrument is running and has been adjusted to section 7-12 of ASTM-B430.

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4.5.6.2.1 Carrier density/viscosity. If sedisperse A12 is used as a carrier, the carrier density and viscosity is supplied on the bottle label by Micromeritics for one degree temperature changes in a range of 24-36 degrees centigrade. If another carrier is used, then the user must establish these temperature dependent physical properties, before measurement.

4.5.6.2.2 Sedimentation rates. Sedimentation rates may be calculated using procedure outlined in MIL-M-382C Amendment 1 under 4.5.11.2 with these exceptions:

P_o = density of sedisperse A12 or another compatible fluid.

n = viscosity of sedisperse A12 or another compatible fluid."

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* 6.8 Delete in its entirety.

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6.12 Add the following new subparagraph:

"6.12 Submission of process control description. The process control description for delay increment consolidation shall be prepared in accordance with DI-E-10392 and submitted to the responsible government technical agency for approval."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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